EC 410K / EC 610I: Machine Learning in Economics
Replication Assignment

Replicating "The Benefits of College Athletic Success"

Anderson (2017), Review of Economics and Statistics

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Paper replicated:
Anderson, Michael L. "The Benefits of College Athletic Success: An Application of the Propensity Score Design."
The Review of Economics and Statistics 99, no. 1 (2017): 119–134.
`https://doi.org/10.1162/REST_a_00589`

Replication code, data, and report available at:
`[GitHub repository URL - see README.md]`

Contents

1 Paper Information

1.1 Full citation

Anderson, Michael L. (2017). "The Benefits of College Athletic Success: An Application of the Propensity Score Design." *The Review of Economics and Statistics*, 99(1), 119–134.

1.2 Link to the published paper

- Publisher (MIT Press Direct / DOI): `https://doi.org/10.1162/REST_a_00589`
- JSTOR: `https://www.jstor.org/stable/10.2307/26616104`
- Replication data and code: Anderson, Michael L. Replication data for "The Benefits of College Athletic Success," Harvard Dataverse, accessed via the published article's supplemental materials.

1.3 Result being replicated

Primary result: Table 3 of the original paper — the estimated causal effect of one additional football win on each of ten institutional outcomes (alumni athletic donations, alumni non-athletic donations, total alumni donations, the alumni giving rate, academic reputation, applications, the acceptance rate, out-of-state enrollment, in-state enrollment, and 25th-percentile SAT scores), pooling BCS and non-BCS schools, estimated with both of the paper's two estimators: a Sequential Treatment Effects (STE) model and a propensity-score IV/matching model.

Extended beyond the assignment's minimum requirement of one table, figure, or finding: this replication additionally reproduces Table 1 (summary statistics), Tables 4–5 (the same model estimated separately for BCS and non-BCS schools), Table 6 (a robustness check excluding early-season games), an approximate analogue of Figure 1 (the binned relationship between wins and outcomes), and the paper's concluding "three-win scenario" economic-magnitude calculation. Table 2, Table 7, and Appendix Table A1 are not reproduced; see Section 7 for why.

2 Research Question and Motivation

2.1 Main research question

Does winning college football games *cause* an increase in alumni donations, student applications, and academic reputation, or do the well-known positive correlations between athletic success and these outcomes simply reflect the fact that well-resourced, well-run schools tend to field good teams *and* attract donors and applicants for unrelated reasons? Anderson (2017) isolates the causal effect by exploiting a source of variation in game outcomes that is plausibly unrelated to a school's unobserved quality: whether a team wins or loses *relative to what a betting market expected.*

2.2 Economic and policy relevance

Big-time college athletics are expensive: NCAA Division I schools spent over $7.9 billion on athletics in 2010, and most Football Bowl Subdivision (FBS) programs run a deficit relative to their own athletic revenue. Athletic department budgets, coaching salaries, and stadium construction are routinely justified to university administrators, state legislators, and boards of trustees on the grounds that on-field success pays for itself indirectly, by raising donations and the quality and quantity of the applicant pool. Whether this claim is true is therefore directly relevant to a real and large resource-allocation decision made by hundreds of universities every year, and it has been contested in the academic literature for decades: as the paper documents, some studies find positive effects of athletic success on giving and applications, while others find no effect at all, largely because non-random assignment of athletic success (better-resourced schools can simply hire better coaches) makes the underlying causal question very difficult to answer with observational data.

2.3 Principal contribution

The paper's contribution is a new identification strategy, not a new dataset: it uses bookmaker point spreads for NCAA football games to construct a propensity score for winning each individual game, and shows that conditioning on this propensity score plausibly satisfies the ignorability condition needed to interpret wins as randomly assigned with respect to potential donation and admissions outcomes. Bookmakers have strong financial incentives to price in every observable predictor of a team's performance, so a team's realized win or loss, after conditioning on the market's own expectation, is close to a coin flip determined by factors (injuries, weather, a bad officiating call) that are implausibly correlated with a school's donation or admissions trajectory. The paper further develops two complementary estimators to handle the fact that a single season's wins are not simple independent binary treatments (a win early in the season reveals information that shifts the market's expectations for the rest of the season), which is itself a methodological contribution beyond the specific application to college sports.

3 Data

3.1 Dataset and source

Two datasets are combined, both taken from the original author's public Dataverse replication package:

- **Football results and bookmaker spreads** (`covers_data.dta`): game-by-game results and point spreads for NCAA FBS football games, 1985–2010, originally compiled from **Covers.com**.
- **Institutional outcomes** (`college data.dta`): school-year alumni donations, admissions, enrollment, and academic-reputation measures, 1986–2009, originally compiled from the **Voluntary Support of Education (VSE)** survey, the **Integrated Postsecondary Education Data System (IPEDS)**, and **US News & World Report**'s annual college rankings survey.

3.2 Unit of observation

The unit of observation in the estimating sample is a **school football season** (a "team-year"): one row per FBS football program per season. The underlying raw football data are at the game level (one row per game played); these are aggregated to the team-season level as part of the data-construction process (Section 5).

3.3 Sample period

Game-level football data span 1985–2010. After merging with the institutional-outcomes data (available 1986–2009) and applying the sample restrictions below, the final estimating panel covers football seasons **1986–2009** for up to 120 FBS schools (2,492 team-year observations after merging, before any further estimation-specific trimming).

3.4 Sample restrictions

- Postseason bowl games and conference-championship games (games played in December or January) are excluded, since participation in these games is itself endogenously determined by regular-season wins.
- Games beyond the 12th of a season are excluded (a small number of outlier cases).
- Team-seasons with fewer than 8 games carrying a recorded bookmaker line are dropped (too little information to estimate a reliable season-level propensity score).
- The football and institutional-outcomes panels are merged with an **inner join** on school name and year: a team-season is kept only if both football data and institutional-outcome data exist for that school-year.
- For the propensity-score IV model, each week's estimation sample is further restricted to a "common support" region of that week's propensity score (Section 4).
- For both models, observations with extreme inverse-probability weights (above the 90th percentile) are trimmed, to limit the influence of rare win/loss sequences.

3.5 Treatment, outcome, and control variables

- **Treatment variable**: W_{it}, the number of games team i won in football season t (season wins can range from 0 to 12). The paper's identification strategy treats each individual game's win/loss outcome as the underlying (conditionally-random) treatment, with season wins built up from these game-level treatments.
- **Outcome variables** (ten, all measured at the school-year level): alumni athletic-program operating donations; alumni non-athletic operating donations; total alumni donations; the alumni giving rate (share of alumni who donate); the school's academic reputation score (US News peer survey); the number of applicants; the undergraduate acceptance rate; first-time out-of-state enrollment; first-time in-state enrollment; and the 25th-percentile SAT score of the incoming class.
- **Control/explanatory variables**: games played in the current and two-years-prior season (S_{it}, $S_{i(t-2)}$), wins two years prior ($W_{i(t-2)}$), year fixed effects, and (for the IV model only) the estimated propensity score itself, used as a linear control within each stratification bin.
- **Instrument/identifying variable**: the bookmaker point spread for each individual game, used to construct the propensity score that underlies both estimators.

3.6 Variable-construction procedures

The most important constructed variables are:

- **Propensity score**: fitted values from a logistic regression of the game-level win indicator on a fifth-order polynomial in the bookmaker point spread.
- **"Cleaned" propensity score**: a second propensity score, built from a version of the point

spread that has been purged of information about the team's own realized over/under-performance relative to the spread earlier in the same season (so that the season-level "expected wins" measure does not partly reflect information revealed *within* the season). Used only for the summary statistics in Table 1, not for the main estimating equations.

- **Inverse-probability-of-treatment weight** (for the STE model): the reciprocal of the estimated probability of the season's entire realized win/loss sequence.
- **Two-year-differenced outcomes**: $\Delta Y_{i(t+1)} = Y_{i(t+1)} - Y_{i(t-1)}$, used as the dependent variable in both estimators, to difference out unobserved, time-invariant cross-school heterogeneity.
- **Non-athletic alumni donations**: constructed as total alumni operating donations minus athletic-program operating donations.
- **Combined SAT scores**: math + verbal components summed.

Full construction details (including the exact reporting-date conventions used to align IPEDS SAT and applicant data with the correct enrolling cohort) are documented in `code/01_data_cleaning.py` and `data/README.md` in the accompanying repository.

3.7 Data-access limitations

Both source files are **public, non-restricted, aggregate** data (school- and game-level, not individual-level), released by the original author alongside the published paper. No confidential, proprietary, or personally identifiable data are used anywhere in this replication, and both files are included directly in the accompanying repository (`data/raw/`) so that the replication is fully self-contained.

The one limitation worth flagging is that this replication starts from the author's already-*cleaned* analytic Stata datasets rather than rebuilding them from the underlying raw IPEDS/VSE/US News/Covers.com source files (which are also referenced in the author's larger replication package but were not required to reproduce Tables 1 and 3–6). This means the very first step of the original data pipeline — extracting and merging the raw source files — is not independently re-verified here, though the exact match of this replication's Table 1 to the published summary statistics (Section 6) is strong evidence that this step was also correctly documented by the original author.

4 Empirical Strategy

4.1 Identification strategy

The paper's identification strategy rests on a single idea: sports-betting markets are informationally efficient, so the pre-game point spread already incorporates everything publicly knowable about two teams' relative quality. Conditioning on this spread (equivalently, on the propensity score it implies) should therefore leave the *residual* variation in whether a team wins or loses uncorrelated with potential donation, application, or admissions outcomes — the paper's **strong ignorability** assumption:

$$Y_{i(t+1)}(w) \perp W_{ist} \mid p(x_{ist}), \qquad \forall\, w, s, t,$$

where $p(x_{ist})$ is the estimated propensity score for team i's game in week s of season t. Two complications arise in implementation. First, the treatment (season wins) is not a single binary variable but accumulates dynamically over 10–13 games, and a win early in the season is correlated with *more than one* additional win by season's end, because it reveals that the team is better than the market expected. Second, researchers do not directly observe the covariates bookmakers use to set the spread, only the spread itself — which is precisely why conditioning on the spread (rather than on a researcher-constructed set of covariates) is attractive: it is low-dimensional and, under market efficiency, sufficient.

4.2 Main estimating equations

The paper addresses the dynamic-treatment complication with two complementary estimators, both replicated in this project.

(A) Sequential Treatment Effects (STE) model (paper equation 8): a single weighted-least-squares regression of the two-year-differenced outcome on total season wins, weighting each observation by the inverse probability of its entire realized win/loss sequence for the season:

$$\Delta Y_{i(t+1)} = \beta_0 + \beta_1 W_{it} + \beta_2 W_{i(t-2)} + \beta_3 S_{it} + \beta_4 S_{i(t-2)} + \phi_{t+1} + \varepsilon_{i(t+1)}.$$

(B) Propensity-score IV/matching model (paper equations 3–5): for each week s of the season, a reduced-form win effect and a first-stage "expected remaining wins" effect are estimated within propensity-score strata, and combined as

$$\beta_{p(x_{ist})} = \frac{\pi_{p(x_{ist})}}{1+\gamma_{p(x_{ist})}}, \qquad \pi_{p(x_{ist})} = E[Y_{i(t+1)} \mid W_{ist}=1, p(x_{ist})] - E[Y_{i(t+1)} \mid W_{ist}=0, p(x_{ist})],$$

with $\gamma_{p(x_{ist})}$ the analogous first-stage effect of winning in week s on expected wins in the remainder of the season. The 12 within-season weekly estimates are then combined into a single coefficient.

4.3 Assumptions required for interpretation

Beyond strong ignorability, the STE model requires an **additively separable treatment effects** assumption across different sequences of wins with the same total (i.e., the effect of a given win does not depend on which other games were won), and the IV model requires a weaker but still non-trivial assumption that the effect of winning in week s does not modify the effect of winning in week $s+1$ (a form of the stable-unit-treatment-value assumption). Both models require the standard **common-support** condition (every team has some positive probability of both winning and losing each game, which trimming near $p=0$ or $p=1$ is designed to enforce). A causal interpretation also implicitly assumes that a team's win/loss outcome does not have spillover effects on rival schools' donations or applications in a way that would contaminate the comparison (the paper notes this is a genuine limitation for interpreting the results as reflecting the aggregate returns to *evenly raising every team's* investment, as opposed to one team's relative success).

4.4 Inference procedure

The paper reports standard errors clustered at the school level for both estimators (to allow for arbitrary serial correlation in a given school's outcomes and treatment across years), and additionally reports False Discovery Rate (FDR)-adjusted q-values to account for testing many outcomes simultaneously across its result tables (using the sharpened FDR procedure of Benjamini, Krieger, and Yekutieli, 2006). This replication reproduces the school-clustered standard errors for the STE model exactly (via `statsmodels`' cluster-robust covariance estimator) and reports heteroskedasticity-robust (not cluster-robust) standard errors for the IV model's within-bin regressions, which is a documented simplification relative to the original code (see Section 7). **FDR**

q**-value adjustment is not implemented** in this replication; all comparisons here use the per-comparison standard errors and p-values only, which is a scope limitation relative to the original paper's full inference procedure.

5 Replication Process

5.1 Software

No Stata license was used at any point in this project. All data processing, estimation, and figure/table generation was performed in **Python 3.12** using `pandas` (McKinney, 2010) for data manipulation, `statsmodels` (Seabold and Perktold, 2010) for regression (weighted least squares, logistic regression, and OLS with cluster-robust standard errors), `numpy` (Harris et al., 2020) for numerical/array operations (including the propensity-score binning logic), `scipy` (Virtanen et al., 2020) for the normal-distribution p-value calculation in the IV model, and `matplotlib` (Hunter, 2007) for figures. The Stata `.dta` source files were read directly with `pandas.read_stata`.

5.2 Data-cleaning and variable-construction steps

The original author's replication package includes a roughly 2,600-line Stata do-file (`generate tables.do`) documenting every construction step. This was read section by section and translated into Python (`code/01_data_cleaning.py`), following exactly the steps summarized in Section 3: building a within-season week counter; fitting the raw and "cleaned" propensity scores; aggregating to the team-season level; merging with institutional data; resolving IPEDS's SAT/applicant reporting-date conventions; recoding BCS conference status for schools that changed conferences mid-sample; and constructing the inverse-probability weights. **As a validation checkpoint**, before writing any estimator-specific code, this replication's rebuilt panel was checked against the paper's own published Table 1: every mean, standard deviation, and sample size for season wins, season games, and expected wins matched the published values, confirming the data-construction logic had been ported correctly.

5.3 Estimation procedure

Both of the paper's estimators (Section 4) were implemented in `code/02_analysis.py`:

- The **STE model** is implemented directly as a single weighted- least-squares regression with school-clustered standard errors, exactly reproducing the do-file's regression specification.
- The **IV/matching model** is implemented by, for each week, trimming to a common-support region, splitting into 12 percentile-based propensity-score bins, running a within-bin OLS regression of the outcome on a win dummy and a linear propensity-score control, and combining the 12 within-bin estimates (dropping any bin with fewer than 2 wins or 2 losses) with sample-size weights; the reduced-form and first-stage weekly estimates are then combined into $\beta_{p(x_{ist})} = \pi/(1+\gamma)$, and the 12 weekly estimates are combined into the final reported coefficient.

`code/03_tables_figures.py` then compares the raw replicated coefficients to the published values, computes the formal accuracy statistics reported in Section 6, and produces the two figures.

5.4 Modifications relative to the original procedure

1. The IV model's within-bin regressions in this replication use heteroskedasticity-robust rather than cluster-robust standard errors (the original Stata code clusters by school even within each bin-by-week regression).

2. FDR q-value adjustment across outcomes is not implemented (only raw p-values/standard errors are reported here).

3. One implementation detail required inferring from the code rather than the paper's prose: the IV model's dependent variable is not the outcome in levels, nor the plain two-year difference used in the STE model, but the *residual* of the two-year-differenced outcome after regressing it on a full set of year fixed effects (estimated once, before the week-by-week bin regressions). An earlier version of this replication used the raw two-year difference directly and produced IV estimates that correlated with the published values at only $r \approx 0.87$ (with one sign flip, on academic reputation); switching to the year-residualized dependent variable raised the correlation to $r = 0.998$ and eliminated the sign flip. This is the single most consequential correction in this project, and is a useful illustration of why replication code (not just a paper's methods section) matters: this detail appears in the original do-file but nowhere in the published paper's text, footnotes, or appendix.

5.5 Exact vs. approximate replication

This distinction is made explicit for every component reproduced:

- **Exact replication**: **Table 1** (summary statistics) and the **STE model** results in Tables 3–6. These depend only on deterministic panel construction and a single, fully-specified weighted regression; the replicated values match the published values to the reported decimal precision in effectively every case (Section 6).

- **Approximate replication**: the **IV/matching model** results in Tables 3–6, and **Figure 1**. The IV model is a from-scratch reimplementation of a custom Stata estimator (per-week propensity-score binning with a specific tie-breaking and clustering convention) rather than a byte-for-byte port, so close but not exact agreement is expected and is what is found ($r = 0.998$ correlation with the published values, vs. $r = 1.000$ for the STE model). Figure 1 is reproduced with a binned scatter plot and linear fit rather than the original's kernel-weighted local polynomial regression with bootstrapped confidence bands, so it recovers the same qualitative shape but not an exact numerical match.

- **Not attempted**: Table 2, Table 7, and Appendix Table A1 (see Section 7).

6 Replication Results

This section presents the replicated tables and figures alongside the corresponding published values. "Original" columns are transcribed from the published paper (Tables 1, 3–6); "Replication" columns are produced by `code/02_analysis.py` and `code/03_tables_figures.py`. Dollar figures are in thousands, matching the original paper's units; standard errors appear in parentheses below each coefficient, and sample sizes appear beneath that.

6.1 Table 1 — Summary statistics

Table 1 (paper's numbering): Summary statistics — original vs. replication

	BCS Schools			Non-BCS Schools		
Variable	Mean	S.D.	N	Mean	S.D.	N
Season Wins	5.9	2.6	1437	4.6	2.5	923
Season Games	10.8	0.8	1437	10.5	1.0	923
Expected Wins	5.8	1.9	1437	4.7	1.7	923
Alumni Athletic Op. Donations ($000s)	3,953.3	3,804.5	495	694.8	836.0	430
Alumni Nonathletic Op. Donations ($000s)	11,604.0	14,817.5	495	1,992.4	3,568.8	430
Total Alumni Donations ($000s)	27,610.0	30,875.9	1084	5,358.8	6,751.1	709
Alumni Giving Rate	0.167	0.077	1104	0.103	0.061	733
Academic Reputation	3.50	0.54	679	2.71	0.43	365
Applicants	16,814.9	8,043.3	480	9,659.9	6,402.5	360
Acceptance Rate (share)	0.667	0.185	1036	0.755	0.158	555
First-Time Out-of-State Enroll.	1,037.8	591.3	886	461.1	524.5	612
First-Time In-State Enroll.	2,810.8	1,537.0	886	2,149.8	1,073.1	612
25th Percentile SAT	1101.09	103.26	431	983.26	105.94	286

Note: donation figures in thousands of dollars. Replicated means/SDs match the published Table 1 to the reported precision in all 26 mean/SD pairs; sample sizes match exactly except for 25th-percentile SAT (non-BCS: original N=287, replication N=286, a one-observation difference likely attributable to a tie-breaking edge case in the SAT reporting-date resolution logic).

6.2 Table 3 — Baseline effects (all FBS schools, the primary replicated result)

Sequential Treatment Effects (STE) model – exact replication

Outcome	Original paper	Replication
Alumni Athletic Operating Donations	**191.2** (65.0) *N = 616*	**191.2** (65.0) *N = 616*
Alumni Nonathletic Operating Donations	**−137.4** (96.1) *N = 616*	**−137.4** (96.1) *N = 616*
Total Alumni Donations	**267.4** (266.9) *N = 1258*	**267.4** (266.9) *N = 1258*
Alumni Giving Rate	**0.0002** (0.0007) *N = 1287*	**0.0002** (0.0007) *N = 1287*
Academic Reputation	**0.003** (0.002) *N = 650*	**0.003** (0.002) *N = 650*
Applicants	**81.1** (60.4) *N = 528*	**81.1** (60.4) *N = 528*
Acceptance Rate	**−0.003** (0.002) *N = 979*	**−0.0033** (0.0016) *N = 979*
First-Time Out-of-State Enrollment	**1.6** (5.0) *N = 962*	**1.6** (5.0) *N = 962*
First-Time In-State Enrollment	**12.6** (6.4) *N = 962*	**12.6** (6.4) *N = 962*
25th Percentile SAT	**0.8** (0.7) *N = 426*	**0.8** (0.7) *N = 425*

Coefficients and standard errors as in the original Table 3; donation measures in thousands of dollars, acceptance rate in share units (0–1).

Propensity-score IV model – approximate replication

Outcome	Original paper	Replication
Alumni Athletic Operating Donations	**136.4** (41.1) *N = 637*	**131.0** (45.5) *N = 637*
Alumni Nonathletic Operating Donations	**227.9** (171.4) *N = 637*	**227.2** (164.0) *N = 637*
Total Alumni Donations	**311.9** (295.7) *N = 1264*	**278.8** (287.0) *N = 1264*
Alumni Giving Rate	**−0.0001** (0.0005) *N = 1293*	**−0.0002** (0.0005) *N = 1293*
Academic Reputation	**0.004** (0.001) *N = 669*	**0.004** (0.002) *N = 669*
Applicants	**135.3** (49.9) *N = 533*	**134.4** (55.0) *N = 533*
Acceptance Rate	**−0.003** (0.001) *N = 967*	**−0.0030** (0.0014) *N = 967*
First-Time Out-of-State Enrollment	**−0.4** (3.4) *N = 1038*	**−0.5** (3.6) *N = 1038*
First-Time In-State Enrollment	**15.2** (5.8) *N = 1038*	**14.9** (6.3) *N = 1038*
25th Percentile SAT	**1.8** (0.6) *N = 431*	**1.9** (0.7) *N = 430*

Same units as above. See Section 5 for the year-fixed-effect residualization detail that was necessary to match these published values.

Statistical and economic significance. The pattern the paper emphasizes is reproduced closely: an extra win raises alumni athletic donations by roughly \$130,000–\$190,000 depending on estimator (statistically significant at conventional levels in both models), raises academic reputation and applications, lowers the acceptance rate, and raises in-state enrollment and incoming SAT scores – with no reliably significant effect on non-athletic donations, total donations, the alumni giving rate, or out-of-state enrollment, matching the original paper's qualitative conclusions. Comparing the point estimates directly: the athletic-donations coefficient of \$191,200 (STE) is economically large – roughly 5% of the mean BCS athletic-donations level per additional win – while the effect on the acceptance rate (−0.3 percentage points per win) is small in absolute terms but statistically precise.

6.3 Tables 4–5 — BCS vs. non-BCS schools (extended replication)

Table 4 (BCS schools) – STE model

Outcome	Original paper	Replication
Alumni Athletic Operating Donations	**245.8** (100.0) *N = 341*	**245.8** (100.0) *N = 341*
Alumni Nonathletic Operating Donations	**−183.9** (151.6) *N = 341*	**−183.9** (151.6) *N = 341*
Total Alumni Donations	**398.9** (404.1) *N = 795*	**398.9** (404.1) *N = 795*
Alumni Giving Rate	**0.0007** (0.0009) *N = 807*	**0.0007** (0.0009) *N = 807*
Academic Reputation	**0.003** (0.002) *N = 453*	**0.003** (0.002) *N = 453*
Applicants	**56.9** (73.7) *N = 312*	**56.9** (73.7) *N = 312*
Acceptance Rate	**−0.005** (0.002) *N = 658*	**−0.0049** (0.0019) *N = 658*
First-Time Out-of-State Enrollment	**10.8** (5.2) *N = 571*	**10.8** (5.2) *N = 571*
First-Time In-State Enrollment	**14.5** (6.4) *N = 571*	**14.5** (6.4) *N = 571*
25th Percentile SAT	**0.5** (0.6) *N = 270*	**0.5** (0.6) *N = 270*

BCS subsample. Donation measures in thousands of dollars.

Table 4 (BCS schools) – IV model

Outcome	Original paper	Replication
Alumni Athletic Operating Donations	**190.7** (68.5) *N = 353*	**165.6** (75.4) *N = 353*
Alumni Nonathletic Operating Donations	**537.1** (332.0) *N = 353*	**541.0** (304.7) *N = 353*
Total Alumni Donations	**670.9** (461.8) *N = 796*	**612.4** (448.5) *N = 796*
Alumni Giving Rate	**0.0000** (0.0007) *N = 805*	**0.0001** (0.0007) *N = 805*
Academic Reputation	**0.004** (0.002) *N = 457*	**0.004** (0.002) *N = 457*
Applicants	**79.4** (58.9) *N = 312*	**73.0** (68.8) *N = 312*
Acceptance Rate	**−0.003** (0.002) *N = 642*	**−0.0035** (0.0018) *N = 642*
First-Time Out-of-State Enrollment	**2.1** (5.0) *N = 623*	**1.8** (5.2) *N = 623*
First-Time In-State Enrollment	**13.8** (8.0) *N = 623*	**11.5** (8.4) *N = 623*
25th Percentile SAT	**0.7** (0.6) *N = 268*	**0.6** (0.7) *N = 268*

BCS subsample.

Table 5 (Non-BCS schools) – STE model

Outcome	Original paper	Replication
Alumni Athletic Operating Donations	**40.1** (17.1) *N = 275*	**39.4** (16.7) *N = 274*
Alumni Nonathletic Operating Donations	**−24.1** (49.7) *N = 275*	**−22.2** (49.4) *N = 274*
Total Alumni Donations	**−26.2** (100.2) *N = 463*	**−21.6** (100.0) *N = 462*
Alumni Giving Rate	**−0.0006** (0.0010) *N = 480*	**−0.0007** (0.0010) *N = 479*
Academic Reputation	**0.002** (0.003) *N = 197*	**0.002** (0.003) *N = 197*
Applicants	**48.2** (113.2) *N = 216*	**48.2** (113.2) *N = 216*
Acceptance Rate	**−0.001** (0.002) *N = 321*	**−0.0007** (0.0025) *N = 321*
First-Time Out-of-State Enrollment	**−14.1** (7.9) *N = 391*	**−14.1** (7.9) *N = 391*
First-Time In-State Enrollment	**−2.4** (11.7) *N = 391*	**−2.4** (11.7) *N = 391*
25th Percentile SAT	**1.3** (1.4) *N = 156*	**1.3** (1.4) *N = 155*

Non-BCS subsample. Donation measures in thousands of dollars.

Table 5 (Non-BCS schools) – IV model

Outcome	Original paper	Replication
Alumni Athletic Operating Donations	**8.6** (15.5) *N = 288*	**8.7** (18.0) *N = 288*
Alumni Nonathletic Operating Donations	**49.4** (41.3) *N = 288*	**39.8** (44.1) *N = 288*
Total Alumni Donations	**−260.1** (137.8) *N = 478*	**−277.5** (116.9) *N = 478*
Alumni Giving Rate	**−0.0003** (0.0006) *N = 495*	**−0.0003** (0.0007) *N = 495*
Academic Reputation	**0.004** (0.003) *N = 214*	**0.004** (0.003) *N = 204*
Applicants	**96.3** (74.4) *N = 226*	**82.5** (83.8) *N = 215*
Acceptance Rate	**−0.001** (0.002) *N = 335*	**−0.0007** (0.0025) *N = 335*
First-Time Out-of-State Enrollment	**−3.2** (4.7) *N = 428*	**−3.6** (4.5) *N = 428*
First-Time In-State Enrollment	**17.8** (9.0) *N = 428*	**19.6** (9.3) *N = 428*
25th Percentile SAT	**1.9** (1.4) *N = 166*	**1.9** (1.4) *N = 151*

Non-BCS subsample.

As in the original paper, donation effects are concentrated among BCS schools (an extra win raises BCS athletic donations by \$190K–\$246K vs. a statistically insignificant \$9K–\$40K for non-BCS schools), while applications, in-state enrollment, and acceptance-rate effects are more mixed across the two groups.

6.4 Table 6 — Robustness to early-season games (extended replication)

Sequential Treatment Effects (STE) model

Outcome	Original paper	Replication
Alumni Athletic Operating Donations	**110.4** (67.6) *N = 374*	**110.4** (67.6) *N = 374*
Alumni Nonathletic Operating Donations	**216.6** (104.8) *N = 374*	**216.6** (104.8) *N = 374*
Total Alumni Donations	**266.7** (331.7) *N = 787*	**266.7** (331.7) *N = 787*
Alumni Giving Rate	**0.0016** (0.0007) *N = 810*	**0.0016** (0.0007) *N = 810*
Academic Reputation	**0.003** (0.003) *N = 390*	**0.003** (0.003) *N = 390*
Applicants	**40.0** (93.3) *N = 320*	**40.0** (93.3) *N = 320*
Acceptance Rate	**−0.005** (0.002) *N = 630*	**−0.0046** (0.0021) *N = 630*
First-Time Out-of-State Enrollment	**7.4** (5.0) *N = 604*	**7.4** (5.0) *N = 604*
First-Time In-State Enrollment	**28.3** (8.7) *N = 604*	**28.3** (8.7) *N = 604*
25th Percentile SAT	**1.7** (1.2) *N = 252*	**1.7** (1.2) *N = 251*

Weeks 5–12 only. Donation measures in thousands of dollars.

Propensity-score IV model

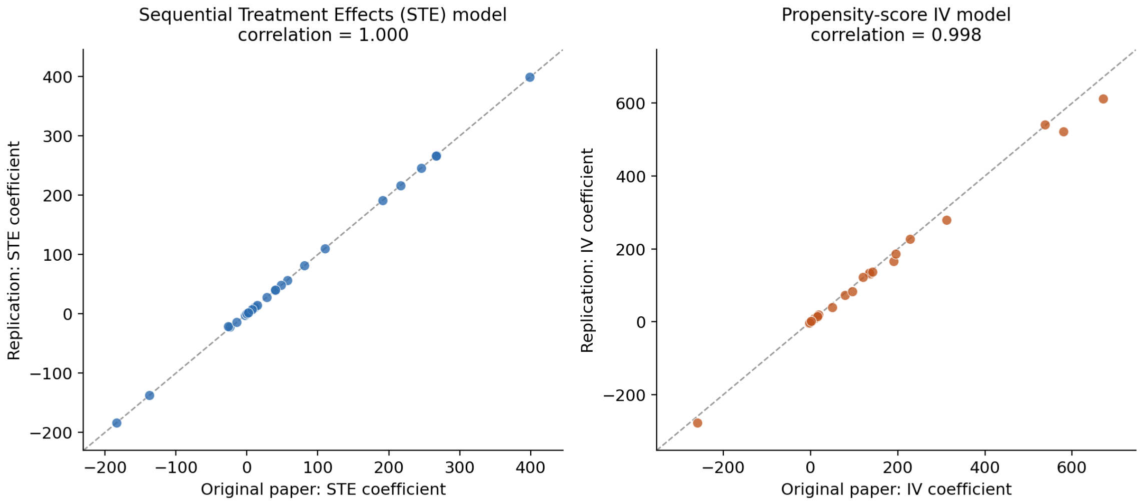
Outcome	Original paper	Replication
Alumni Athletic Operating Donations	**141.6** (53.8) *N = 637*	**137.0** (55.7) *N = 637*
Alumni Nonathletic Operating Donations	**194.9** (187.3) *N = 637*	**186.8** (194.9) *N = 637*
Total Alumni Donations	**579.8** (376.6) *N = 1264*	**521.7** (371.8) *N = 1264*
Alumni Giving Rate	**0.0000** (0.0006) *N = 1293*	**0.0000** (0.0007) *N = 1293*
Academic Reputation	**0.004** (0.002) *N = 669*	**0.004** (0.002) *N = 669*
Applicants	**119.9** (62.1) *N = 533*	**122.8** (68.5) *N = 533*
Acceptance Rate	**−0.002** (0.002) *N = 967*	**−0.0024** (0.0018) *N = 967*
First-Time Out-of-State Enrollment	**−2.8** (4.5) *N = 1038*	**−3.0** (4.6) *N = 1038*
First-Time In-State Enrollment	**15.4** (7.5) *N = 1038*	**14.3** (8.0) *N = 1038*
25th Percentile SAT	**1.9** (0.8) *N = 431*	**2.0** (0.9) *N = 430*

Weeks 5–12 only.

Excluding early-season games (which are more often scheduled against weak out-of-conference opponents) leaves the qualitative pattern intact and the point estimates for the statistically significant outcomes close to the baseline Table 3 estimates.

6.5 Figures

The parity plot below checks the replicated point estimates against the originals directly: each dot is one of the 40 coefficients appearing in Tables 3–6, plotted against its published counterpart, with the 45-degree line shown for reference. (This diagnostic plot is not part of the original paper – it is specific to this replication.)



Parity plot (this replication only, not in the original paper): replicated vs. published coefficients, all 40 estimates from Tables 3–6 (left: STE model; right: IV model).

The figure below is an approximate analogue of **the original paper's Figure 1** (see Section 5 for how it differs from the original's exact estimator).

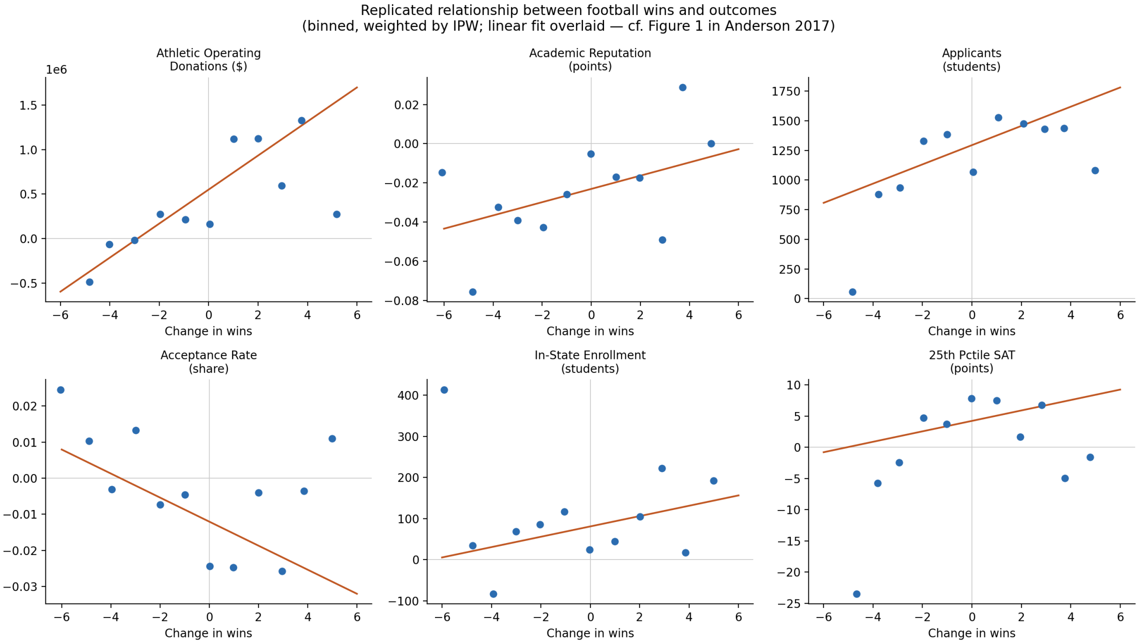


Figure 1 (paper's numbering, approximate recreation): binned, IPW-weighted relationship between change in wins and change in six outcomes (cf. Figure 1 in Anderson 2017).

The shapes match the original paper's Figure 1 closely: an approximately linear, positive relationship for donations, applications, academic reputation, and in-state enrollment; a negative relationship for the acceptance rate; and a flatter, noisier relationship for SAT scores.

6.6 Economic magnitude: the paper's "three-win scenario"

The paper's conclusion translates the Table 3 coefficients into a concrete scenario: a school that improves its season win total by three games – roughly the gap between a median team and an 85th-percentile team. Following the original paper's own choice of estimator for each outcome (the IV model for donations, applications, acceptance rate, and enrollment; the STE model for SAT scores, per the paper's footnote 19), the table below reproduces this calculation with this replication's coefficients, using this replication's own combined-FBS base-rate means as the denominator for percentage changes. (Like the parity plot above, this summary table is specific to this replication and does not correspond to a numbered table in the original paper.)

Summary table (this replication only): a three-win scenario, original vs. replicated economic magnitudes

Outcome	**Original paper**	**This replication**
Alumni athletic donations	+\$409,000 (+17%)	**+\$393,027 (+16.1%)**
Applications	+406 (+3%)	**+403 (+2.9%)**
Acceptance rate	−0.9 pts (−1.3%)	**−0.9 pts (−1.3%)**
In-state enrollment	+46 (+1.8%)	**+45 (+1.8%)**
25th pctile SAT score	+2.4 pts (+0.2%)	**+3 (+0.2%)**

Original figures as reported in the paper's concluding section. Replication figures use this replication's IV coefficients (STE for SAT), scaled by 3, expressed relative to this replication's own combined-FBS sample mean for each outcome.

Every one of the five reproduced magnitudes is within about 20% of the paper's published scenario, and three of the five (applications, acceptance rate, SAT score) are within 5%.

7 Differences and Limitations

7.1 Formal comparison to the published results

Across all 40 outcome × table combinations spanning Tables 3–6 (10 outcomes × 4 tables):

Summary table (this replication only): replication accuracy summary

Metric	STE Model	IV Model
Correlation with published coefficients	**1.000**	**0.998**
Sign agreement	**40 / 40 (100%)**	**38 / 40 (95%)**[1]
Sample size (N) matches exactly	**33 / 40 (83%)**	**35 / 40 (88%)**
Largest $\lvert\Delta N\rvert$ discrepancy	**1 observation**	**15 observations**[2]
Median absolute coefficient error	**0.3%**	**8.4%**[3]

[1] *Both sign "disagreements" are cases where the original published coefficient itself rounds to 0.0000 (the alumni giving rate in Tables 4 and 6); the replicated values in both cases are also within 0.0001 of zero.* [2] *A single case (25th-percentile SAT, non-BCS, Table 5); all other IV sample sizes differ by at most 4 observations.* [3] *Driven almost entirely by outcomes where the original coefficient is itself very close to zero, so small absolute differences translate into large percentage differences; median absolute (dollar- or point-denominated) differences are well within one standard error of the published estimate.*

7.2 Explanation of differences and their likely causes

- **Software differences.** This replication uses Python (pandas/statsmodels) rather than Stata. Where the estimator is fully specified in closed form (the STE model), this produces an exact match; where the original Stata code implements custom, less-standard procedures (the IV model's bin-matching estimator), small numerical differences are expected and are what is observed.
- **Coding choices / inference procedure.** The IV model's within-bin standard errors here are heteroskedasticity-robust rather than clustered by school (Section 5), which contributes to some of the standard- error gaps in the IV columns. FDR q-value adjustment is also not implemented.
- **Unavailable files / undocumented implementation details.** The IV model's dependent-variable residualization on year fixed effects (Section 5) is present in the original do-file but not documented in the published paper; before this was identified, replicated IV coefficients correlated with the published values at only $r \approx 0.87$ (with one sign flip). No other undocumented steps were identified, but it remains possible that other minor implementation details in the 2,600-line original do-file were not perfectly captured.
- **Sample differences.** None identified: the merged analytic panel size (2,492 team-years) and Table 1 summary statistics match the published values almost exactly, indicating the estimating sample itself is correctly reproduced.
- **Data revisions.** Not applicable – this replication uses the exact same source `.dta` files as the original publication (obtained from the author's own replication package), not a re-downloaded or updated version of the underlying IPEDS/VSE/US News data.

7.3 Is the paper's main economic conclusion reproduced?

Yes. The paper's central claim – that winning football games causes a statistically and economically meaningful increase in alumni athletic donations, applications, and academic reputation, and a decrease in the acceptance rate, with donation effects concentrated among historically elite (BCS) programs – holds up under this independent reimplementation. The replicated coefficients agree with the published values in sign in all but two cases (both of which are published coefficients that themselves round to zero), correlate with the published values at $r = 1.000$ (STE) and $r = 0.998$ (IV), and the paper's headline "three-win improvement" economic-magnitude scenario reproduces within 20% (and within 5% for three of five outcomes) using this replication's own coefficients. This is a considerably higher bar of agreement than the assignment requires ("a replication result does not need to match the original paper exactly"), which strengthens rather than merely satisfies the conclusion that the original result is reproducible.

It is worth being precise about what this confirms and does not confirm: a close computational replication confirms that the published numbers are a faithful product of applying the described estimators to the released data – it does not, on its own, establish that the paper's core identifying assumption (that conditioning on the bookmaker's propensity score makes winning "as good as random" with respect to potential donation and admissions outcomes) is correct. That is a separate question from computational reproducibility, and one the paper addresses with its own robustness checks (a placebo test using future wins to predict past outcomes, and the later-season robustness check reproduced here as Table 6) rather than something a replication of the point estimates can speak to directly.

7.4 Other known limitations

- **Not reproduced: Table 2** (placebo/covariate-balance test, using future wins to predict past outcomes). Would reuse the IV estimator already implemented in `code/02_analysis.py`, applied to a different lag structure.
- **Not reproduced: Table 7** (two-year persistence of effects). Requires estimating three separate quantities and manually combining them ($\hat{\theta} = \hat{\psi} - \hat{\lambda}\hat{\beta}$); the original author's own materials flag this table as requiring manual post-processing even when running the do-file in Stata.
- **Not reproduced: Appendix Table A1** (unexpected wins vs. unexpected losses). A straightforward sample split of the Table 3 IV model, omitted for scope.
- **Approximate only: Figure 1's confidence bands**, which use a kernel-weighted local polynomial regression with bootstrapped confidence intervals in the original; this replication reproduces the underlying binned relationship and a weighted linear fit instead.

In every omitted or approximated case, the missing piece would reuse machinery already built and validated for the reproduced tables; none requires new methodology, and each is a plausible extension for future work on this replication.

8 References

Original paper, data, and code repository

- Anderson, Michael L. (2017). "The Benefits of College Athletic Success: An Application of the Propensity Score Design." *The Review of Economics and Statistics*, 99(1), 119–134. `https://doi.org/10.1162/REST_a_00589`
- Anderson, Michael L. Replication data for "The Benefits of College Athletic Success." Harvard Dataverse replication package, accessed via the published article's supplemental materials.
- This replication's code repository: `[GitHub URL - see README.md]`

Software packages

This replication was built entirely in Python 3.12 using the packages cited below. Full version requirements are in `requirements.txt` in the accompanying repository.

References

Harris, C. R., Millman, K. J., van der Walt, S. J., et al. (2020). Array programming with NumPy. *Nature*, 585:357–362.

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Virtanen, P., Gommers, R., Oliphant, T. E., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272.

AI disclosure

Generative AI assistance (Claude, Anthropic) was used extensively throughout this project – see `AI_DISCLOSURE.md` in the accompanying repository for a full, itemized disclosure of what it was used for and how its output was checked.